Classification of Eye Diseases using Machine Learning with Preprocessing

2024 8th International Artificial Intelligence and Data Processing Symposium (IDAP) | 979-8-3315-3149-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/IDAP64064.2024.10710977

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***Abstract*—The eye, as the sensory organ responsible for vision, is an essential part of our daily lives. However, various disorders can negatively impact eye health and visual acuity, potentially resulting in visual impairment. Machine learning holds immense potential in the diagnosis of eye diseases. This study investigates the utilization of machine learning models for the early detection of ocular diseases. After processing the retinal images with various image processing techniques, distinctive features are extracted using feature extraction algorithms such as histogram of oriented gradients (HOG), local binary patterns (LBP), and residual network-50 (ResNet-50). The obtained training data is applied to the classification algorithms like K-nearest neighbors (KNN), support vector machine (SVM), and extreme gradient boosting (XGBoost) to differentiate between normal eyes and those with diabetic retinopathy, cataracts, and glaucoma. These three different algorithms for classifying the eye diseases are compared according to key performance metrics. The results indicate that the combination of XGBoost with ResNet-50 achieves the highest performance with 92% accuracy, followed by the combination of SVM with ResNet-50 at 90% accuracy.**

Keywords — Machine learning, eye disease, feature extraction, image processing

I. INTRODUCTION

The eyes play a crucial role in fulfilling essential needs in our daily lives, such as reading, driving, recognizing faces, and perceiving the surrounding environment. Eye diseases can significantly diminish the quality of life by causing vision loss or blindness. According to the World Health Organization's (WHO) 2023 publication, 2.2 billion people worldwide are affected by eye disorders [1]. It is estimated that approximately 94 million people worldwide suffer vision loss due to cataracts and 7.7 million due to glaucoma, while the number of individuals experiencing vision loss as a result of diabetic retinopathy is estimated to be 3.9 million. A vast majority of individuals encounter at least one eye condition during their lifetime, necessitating appropriate care and treatment. Early diagnosis and treatment of these conditions can help preserve individuals' eye health and enhance their quality of life by preventing or managing the progression of the disease.

Machine learning-based classification methods enable healthcare centers to accurately diagnosing eye diseases in their early stages and to correctly categorize patients into appropriate risk groups. These methods, which learn from large datasets, improve diagnostic processes by reducing errors associated with human factors and provide more effective support for patients' treatment processes. Ophthalmologists can use machine learning models to analyze patient images and make early diagnoses while also reducing the need for in-person examinations by remotely monitoring patients' eye problems.

Researchers employed machine learning techniques to classify various eye diseases. Utilizing the ocular disease intelligent recognition (ODIR) dataset comprising 5000 diverse fundus images, they aimed to predict the presence of specific eye diseases in patients. In binary classification, the highest accuracy of 75% was achieved by the naive bayes (NB) algorithm, whereas in multi-class classification, the highest accuracy of 88% was attained by the random forest (RF) algorithm [2]. Researchers compared the performance of RF and K-nearest neighbors (KNN) algorithms for classifying eye diseases using image-based features. RF, with 100 decision trees, achieved an accuracy of 80%, outperforming KNN (3 neighbors) with 70% accuracy. This suggests RF's superior ability in handling complex image variations and classifying eye diseases [3]. Researchers compared support vector machine (SVM) and RF methods for the detection of diabetic retinopathy. The RF method achieved an accuracy of 91% on the diaretdb1 dataset and 85% on the messidor dataset, while the SVM method achieved 88% and 80% accuracy, respectively. These results highlight the potential of the RF method for the early diagnosis of diabetic retinopathy. Furthermore, this study suggests that data characteristics and interpretability are important factors in method selection [4].

Aim of this study is to achieve significant advancements in the field of ocular health through the use of machine learning methods. For this purpose, the study focuses on analyzing medical imaging data to enable the early diagnosis and classification of common eye diseases, thereby facilitating the implementation of early treatment methods. Machine learning technologies can be effectively utilized to develop better treatment and management strategies in ocular health, ultimately contributing to the improvement of public health.

This study delves into the application of machine learning for the classification of eye diseases, utilizing a dataset comprising images of normal eyes and those affected by diabetic retinopathy, cataracts, and glaucoma. The study employs image processing techniques like grayscale

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conversion, histogram equalization, and resizing to enhance image quality and optimize algorithm performance. Feature extraction processes are also performed using histogram of oriented gradients (HOG), local binary patterns (LBP) and residual network-50 (ResNet-50) techniques to capture distinctive image characteristics, which are then fed into classifiers like KNN, SVM, and extreme gradient boosting (XGBoost) for disease prediction. The results underscore the superior performance of models trained with image processing and feature extraction, showcasing the efficacy of machine learning in automating eye disease diagnosis.

The rest of this article is organized as follows. In Section II, we provide a detailed overview of the dataset used in the study. In Section III, we elucidate the image processing techniques employed to prepare the images for subsequent analysis. In Section IV, we elaborate on the feature extraction methods utilized to capture distinctive image characteristics crucial for accurate classification. In Section V, we provide a comprehensive description of the machine learning models and methodological approaches adopted for model training, hyperparameter optimization, and performance evaluation. In Section VI, we present the outcomes of the study, showcasing the efficacy of these techniques in enhancing classification accuracy. Finally, the conclusion in Section VII summarizes the key findings, emphasizing the potential of machine learning in automating eye disease diagnosis.

II. Dataset Description

Datasets derived from sensitive medical data hold immense potential for advancing AI-based diagnostic and therapeutic approaches. However, strict adherence to ethical guidelines throughout data collection and utilization processes is paramount. The use of such data necessitates meticulous consideration of ethical concerns, including patient consent, data privacy, and transparency. The dataset we used provides a comprehensive resource for the training and evaluation of machine learning models [5]. The dataset includes approximately 1,000 retinal images for each class of eye disorders, such as cataract, diabetic retinopathy, and glaucoma, along with normal eye data to distinguish between these disorders. This balanced distribution shown in Figure 1 enables the models to learn in a rich and diverse environment, promoting generalization. The dataset's focus on retinal images enables the development of models specifically designed for the accurate diagnosis and classification of eye diseases. The extraction of critical features from retinal images using appropriate methods enhances the ability of machine learning models to detect eye diseases, leading to more reliable results in clinical applications. The quality of the dataset requires the control of factors such as image resolution, contrast, and clarity, as well as the application of preprocessing steps when necessary.

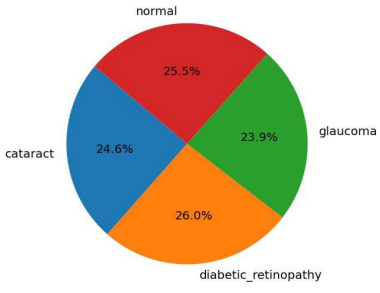


Figure. 1. Dataset distribution

The images in the dataset are presented in various formats and resolutions. Cataract and glaucoma images are provided in 256x256 pixel resolution JPG and PNG formats, while normal eye images are in 512x512 pixel resolution JPG format, and diabetic retinopathy images are in 512x512 pixel resolution JPEG format. This diversity enables the model to adapt to different image conditions and resolutions, helping it to cope more effectively with the varying situations it may encounter in real-world clinical settings.

III. Image Proccessing Procedure

Image processing is the process of analyzing, modifying, and enhancing a digital or analog image using various methods in a computer environment. This process is carried out with the aim of extracting meaningful information from images, improving their quality, removing unwanted elements, or highlighting specific features. The primary goal of image preprocessing is to prepare the image for subsequent analysis. This may involve noise reduction, contrast enhancement, sharpening, color correction, and other operations. Here, we will provide an overview of the image processing techniques employed.

A. Converting an Image to Grayscale Color Space

This transformation converts the image from the red, green and blue (RGB) color space to the grayscale color space as shown in Figure 2. While the RGB color space uses three channels to represent each pixel of the image, the grayscale color space represents the same visual information using only one channel. The purpose of this conversion is to represent the data using less storage space and processing power, which is particularly beneficial in applications with large data processing and speed requirements. A grayscale image is stored as an 8-bit integer value for each pixel, allowing the pixels to represent a grayscale level ranging from 0 (black) to 255 (white) [6].

The following equation is typically used to convert an image from RGB to grayscale:

$$\text{Gray} = 0.3 * R + 0.59 * G + 0.11 * B \quad (1)$$

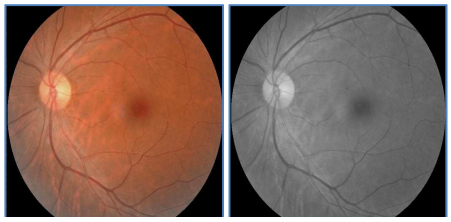


Figure. 2. An RGB retinal image and its conversion to grayscale image.

B. Blurring

In image processing applications, convolution operations using a low-pass filter help blur images, thereby reducing small details and noise. This method removes unnecessary details from the image, making the focused object or important features more prominent.

C. Resizing

All images in the dataset are resized for training purposes due to their varying dimensions. This resizing ensures consistency in input size for the machine learning models and facilitates efficient training and analysis.

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D. Histogram Equalization

Histogram equalization (HE) is a contrast enhancement technique used in image processing. It modifies the distribution of pixel values within an image, thereby increasing the contrast between dark and light regions. This results in a more detailed and vivid image. Additionally, HE is effective at reducing noise and blur. While frequently applied to grayscale images, HE can also be utilized on color images [7].



Figure. 3. A grayscale retinal image and its conversion to histogram equalization image.

When histogram equalization is applied to a grayscale retinal image, the distribution of pixel values spreads over a wider range. Dark regions become brighter, while bright regions become slightly darker. This increases the contrast of the image, revealing more details as shown in Figure 3. The distribution of pixels in the histogram becomes more uniform.

E. Thresholding

Thresholding is a fundamental technique used in image processing. It converts an image into a binary image based on a specified threshold value as shown in Figure 4. Specifically, if the value of each pixel is greater than the threshold, it is assigned a white value (1), while if it is less than the threshold, it is assigned a black value (0) [8].

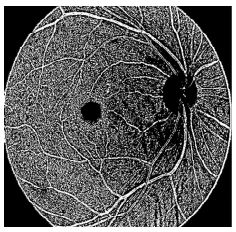


Figure. 4. A retinal image with thresholding.

IV. Feature Extraction

Feature extraction is a dimensionality reduction technique where essential information from a large amount of input data is compressed and transformed into a smaller-sized representation. The primary goal is to discover the most meaningful information within the input data and express it as a lower-dimensional vector. If the input to the algorithm is too large and contains irrelevant details (i.e., high volume but low amount of actual information), this data is transformed into a smaller set of representations called a feature vector. This process is known as feature extraction. When the appropriate features are selected, this feature vector is expected to perform the desired task with this smaller representation instead of the full-sized input data.

A. Feature Extraction with Histogram of Oriented Gradients

Histogram of oriented gradients (HOG) is a feature extraction method used to capture local shape information in an image, and it is widely used in object detection and recognition. HOG represents edge and texture information in an image through gradients (directional changes in pixel intensity) calculated for each pixel in horizontal and vertical directions. The image is then divided into small cells, and the gradient directions within each cell are divided into a fixed number of bins to create a histogram. This histogram represents the dominant edge directions and their magnitudes within the cell. Several cells are grouped together to form blocks, and the histogram values within the blocks are normalized to make them more robust to changes in illumination. Finally, the normalized histogram values from all blocks are combined to form a HOG feature vector that represents the image. This feature vector can be used for tasks such as classification.

B. Feature Extraction with Local Binary Patterns

Local binary patterns (LBP) is a feature extraction method used to describe local texture patterns in an image. It is widely employed in applications such as face recognition, emotion analysis, and texture classification. LBP operates by comparing the intensity of a pixel with its neighboring pixels and converting the resulting binary patterns into decimal numbers. These decimal numbers represent the LBP codes of the pixels, and the frequency distribution of these codes, calculated for all pixels, forms an LBP histogram, which represents the texture features of the image [9].

C. Feature Extraction with Residual Network-50

Residual network (ResNet), a convolutional neural network (CNN) architecture, marks a significant advancement in deep learning, particularly excelling in image classification tasks. The defining characteristic of ResNet is its incorporation of residual connections, which mitigate the vanishing gradient problem often encountered in deep networks. This enables the training of deeper networks, leading to enhanced performance.

ResNet-50, a 50-layer deep neural network architecture, comprises convolutional, pooling, and fully connected layers. The initial layers of the network are responsible for learning low-level features in the image (such as edges and corners), while the deeper layers capture more abstract and high-level features (like object parts and textures). During the feature extraction process, an input image is fed through the ResNet-50 network, with distinct features being extracted at each layer. In the final layer, a feature vector, a condensed representation of the image, is obtained. This feature vector can then be utilized for various tasks, including image classification.

V. Models

A. K-Nearest Neighbors

The K-nearest neighbors (KNN) algorithm is a fundamental and robust supervised learning method employed in addressing conventional machine learning classification problems. Its core principle involves examining the nearest neighbors within an existing dataset to ascertain the class or value of a novel data point. The effectiveness of the KNN algorithm in image classification hinges on the quality of the features utilized, the size of the dataset, and the selection of the K value. Upon the arrival of a new element, the distance

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to its nearest neighbors is computed for the purpose of categorization and classification into distinct containers. The distance is calculated using the Euclidean distance, which represents the shortest distance between two points. In a KNN algorithm characterized by well-chosen features and a sufficiently large dataset, high accuracy rates can be achieved. The operational principle of the KNN algorithm is remarkably straightforward. It obviates the need for intricate parameter adjustments, thereby facilitating its rapid execution. However, in the context of extensive datasets, computational costs can escalate, as distance calculations with the entire dataset are necessitated for each new data point. Furthermore, it exhibits sensitivity to noisy or outlier values within the dataset, underscoring the significance of data preprocessing steps. Its performance may deteriorate in high-dimensional datasets.

In Figure 5, data points belonging to different classes are represented by points of different colors, while the red star represents a new data point that needs to be classified. The KNN algorithm calculates the distances between the new data point and all other data points, and identifies the K nearest neighbors. The value of K is a hyperparameter determined by the user. The class of the new data point is determined by the majority vote of its K nearest neighbors. In the figure, the circle around the red star represents a boundary line used to determine the nearest neighbors for K=7. Since 4 out of the 7 neighbors within this boundary line belong to the green class, the KNN algorithm will classify the new data point as belonging to the green class. On the other hand, for K=3, the red star is classified as belonging to the green class. Although the KNN algorithm is simple, easy to understand, and has a fast training time, it can slow down on large datasets, is sensitive to noise and outliers, and the correct choice of the K value positively affects its performance.

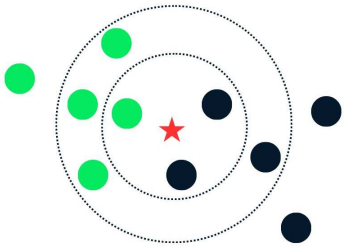


Figure. 5. The working principle of the KNN classifier algorithm [10].

B. *Support Vector Machine*

Support vector machine (SVM) is a powerful supervised learning algorithm used in machine learning to solve classification problems using classical methods. SVM aims to create the widest possible separation between classes by mapping datasets into a high-dimensional space. The data points closest to this separating boundary are called as support vectors. SVM operates with kernel functions that transform data into a higher-dimensional space, making classes that are not linearly separable in the original space separable in this new space. As a result of this transformation, a hyperplane that best separates the classes is determined with the help of the support vectors. SVM classifies new data points based on their position relative to this hyperplane, thereby successfully performing the classification task. This algorithm can be particularly effective in image classification, especially when working with well-defined features and relatively small datasets. SVM can achieve high accuracy rates and is resistant to the problem of overfitting. However, its computational cost can be high with large datasets, and the choice of kernel function can significantly impact performance.

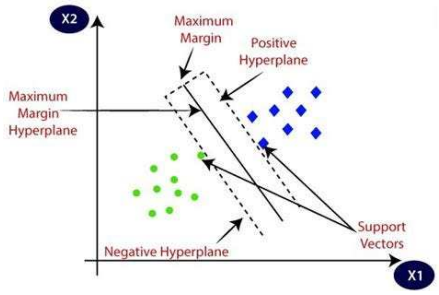


Figure. 6. The working principle of the SVM classifier algorithm [11].

In Figure 6, data points belonging to two different classes (green and blue) are shown in a two-dimensional space. The goal of SVM is to find a hyperplane that best separates these two classes. In two dimensions, a hyperplane is a line, in three dimensions, it is a plane, and in higher dimensions, it is a hypersurface. SVM determines this hyperplane by maximizing the distances to the data points of the classes, a distance known as the margin. In the visualization, the positive hyperplane and negative hyperplane delineate the boundaries of the margin. The distance between these two hyperplanes is called the maximum margin, and the optimization goal of the SVM algorithm is to maximize this margin. support vectors are the data points that define the boundaries of the margin and are closest to the hyperplanes. The SVM algorithm uses only the support vectors to determine the hyperplane; other data points do not affect the classification. This makes SVM more robust against noisy data. The classification performance of the SVM algorithm depends on the width of the margin between the hyperplane and the support vectors. The wider the margin, the better the classification. Therefore, the SVM algorithm seeks to find the optimal hyperplane that maximizes the margin.

C. *Extreme Gradient Boosting*

Extreme gradient boosting (XGBoost) is a powerful machine learning algorithm that operates by combining multiple decision trees to make predictions. These trees are constructed sequentially, with each tree attempting to rectify the errors of its predecessor. This iterative process enhances the model's overall performance. Notably, XGBoost leverages parallel processing capabilities, enabling the simultaneous construction of trees and facilitating rapid execution on large datasets. Additionally, the algorithm employs L1 and L2 regularization techniques to mitigate overfitting. A significant advantage of XGBoost is its automatic handling of missing data, streamlining the data preprocessing phase. Furthermore, tree pruning is utilized to remove underperforming branches, thereby reducing model complexity and improving generalization [12].

VI. Approaches and Results

A machine learning model designed for eye disease classification employs retinal images, meticulously categorized by disease type. Feature extraction is performed using a combination of advanced techniques: a pre-trained ResNet-50 deep neural network captures high-level features, HOG focuses on edge and shape information, while LBP

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extracts texture characteristics. This multi-faceted approach ensures a comprehensive representation of the retinal images. The dataset is then split into train (80%) and test (20%) sets. Three powerful machine learning models KNN, SVM, and XGBoost are trained on the extracted features. To fine-tune performance, hyperparameter optimization techniques like grid search and random search are employed, systematically exploring different parameter combinations. Additionally, cross-validation is utilized to rigorously evaluate each model's generalization ability and mitigate the risk of overfitting. This ensures their capacity to accurately classify new, unseen retinal images, ultimately contributing to the advancement of automated eye disease diagnosis.

In summary, the general flowchart of the classification methods shown in Figure 7 represents the sequential steps followed for classifying a set of images through feature extraction and machine learning. This process encompasses image processing techniques (grayscale conversion, blurring, histogram equalization, thresholding), feature extraction methods (ResNet-50, HOG, LBP), and the training and evaluation of a machine learning model (KNN, SVM, XGBoost). The performance of the models is evaluated according to key performance metrics depending on whether or not the images are preprocessed.

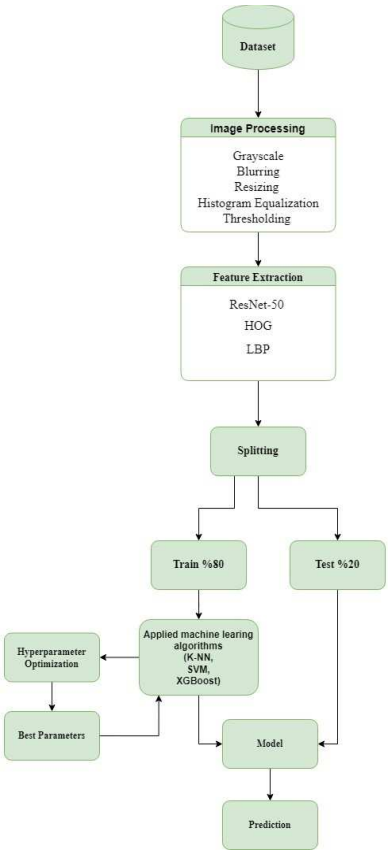


Figure. 7. Flowchart of model architecture for classification methods.

A. *Disease Prediction without Image Processing and Feature Extraction*

The findings distinctly illustrate the differences in performance among the algorithms and underscore the variations in their ability to generalize to new data. By analyzing these results, one can gain a better understanding of how each algorithm performs in practice and how robust they are in diverse scenarios.

TABLE I. PERFORMANCE OF CLASSIFIERS WITHOUT IMAGE PROCESSING AND FEATURE EXTRACTION

Classtification Algorithms	Performance Measures			
	Precision	***Recall***	***F1-score***	***Accuracy***
KNN	69%	69%	68%	70%
SVM	77%	76%	76%	76%
XGBoost	82%	82%	82%	82%

The results presented in Table I compare the performance of different classification algorithms (KNN, SVM, and XGBoost) without employing image processing and feature extraction techniques. XGBoost clearly outperforms the others, achieving the highest scores across all metrics: precision (82%), recall (82%), F1-score (82%), and accuracy (82%). SVM also demonstrates good performance with scores consistently around 76-77%, while KNN exhibits the lowest performance overall. These findings emphasize the potential of XGBoost as a powerful classifier even in the absence of image processing or feature extraction.

B. *Disease Prediction with Image Processing and Feature Extraction*

1) *Disease prediction using HOG and LBP*

In this case, the images are converted from RGB color format to grayscale format and resized to 128x128 pixels. Histogram equalization is applied, followed by blurring, for image enhancement. Thresholding techniques are also employed. Feature extraction is performed on the images using HOG and LBP. During the process, N-fold cross-validation method is employed for different N values (N=1, 2, 3, 4, 5) to assess the discrepancy between test accuracies and to mitigate the potential for overfitting. KNN, SVM, and XGBoost algorithms are chosen for the classification process.

TABLE II. PERFORMANCE OF KNN CLASSIFIER WITH IMAGE PROCESSING, HOG, AND LBP

KNN Classfier	KNN N-Fold Performance Measures			
	Precision	***Recall***	***F1-score***	***Accuracy***
1-Fold	83%	84%	84%	84%
2-Fold	82%	83%	81%	82%
3-Fold	83%	83%	83%	84%
4-Fold	84%	84%	83%	84%
5-Fold	82%	83%	81%	82%

TABLE III. PERFORMANCE OF SVM CLASSIFIER WITH IMAGE PROCESSING, HOG, AND LBP

SVM Classfier	SVM N-Fold Performance Measures			
	Precision	***Recall***	***F1-score***	***Accuracy***
1-Fold	85%	84%	84%	84%
2-Fold	83%	83%	83%	84%
3-Fold	85%	85%	84%	85%
4-Fold	84%	84%	83%	84%
5-Fold	84%	84%	83%	84%

TABLE IV. PERFORMANCE OF XGBOOST CLASSIFIER WITH IMAGE PROCESSING, HOG, AND LBP

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XGBoost Classfier	XGBoost N-Fold Performance Measures			
	Precision	*Recall*	*F1-score*	*Accuracy*
1-Fold	84%	84%	84%	84%
2-Fold	86%	87%	87%	87%
3-Fold	85%	85%	85%	85%
4-Fold	86%	85%	85%	85%
5-Fold	83%	84%	84%	84%

The evaluation of results obtained using image processing and feature extraction techniques (HOG, LBP) indicates a performance improvement compared to Table I. The results in Tables II, III, and V reveals that these techniques positively impact the performance of all classifiers, with the most notable enhancement observed in the KNN algorithm, achieving peformance values of 83-84%. However, this performance falls short than other algorithms. XGBoost appears to have the best performance with 86-87%. SVM shows a performance close to XGBoost.

2) Disease prediction using ResNet-50

In this experiment, images are converted from RGB color format to grayscale format and resized to 244x244 pixels. Feature extraction is performed on the images using the ResNet-50 model, and training is conducted on these features. During the training process, N-fold cross-validation is employed. KNN, SVM, and XGBoost algorithms are used for the classification process.

TABLE V. PERFORMANCE OF KNN CLASSIFIER WITH IMAGE PROCESSING AND RESNET-50

KNN Classfier	KNN N-Fold Performance Measures			
	Precision	*Recall*	*F1-score*	*Accuracy*
1-Fold	86%	85%	86%	85%
2-Fold	84%	84%	84%	84%
3-Fold	84%	85%	85%	84%
4-Fold	85%	85%	85%	85%
5-Fold	87%	87%	87%	87%

TABLE VI. PERFORMANCE OF SVM CLASSIFIER WITH IMAGE PROCESSING AND RESNET-50

SVM Classfier	SVM N-Fold Performance Measures			
	Precision	*Recall*	*F1-score*	*Accuracy*
1-Fold	89%	89%	89%	89%
2-Fold	87%	87%	87%	87%
3-Fold	88%	88%	88%	88%
4-Fold	91%	91%	90%	90%
5-Fold	90%	90%	90%	90%

While the initial experiment results in Table I underscored the superior performance of XGBoost without image processing and feature extraction, subsequent results presented in Table V, Table VI, and Table VII offer a new perspective when these techniques are incorporated. The inclusion of image processing and ResNet-50 led to improvements across all classifiers, with the most notable enhancement observed in KNN, achieving a score of 87% as shown in Table V. This suggests that the selection of the optimal classifier is contingent upon the specific feature extraction and preprocessing methods employed. SVM and XGBoost also exhibited high performance after image processing, attaining performance values of 90-91% and 92% as seen in Table VI and VII, respectively.

TABLE VII. PERFORMANCE OF XGBOOST CLASSIFIER WITH IMAGE PROCESSING AND RESNET-50

XGBoost Classfier	XGBoost N-Fold Performance Measures			
	Precision	*Recall*	*F1-score*	*Accuracy*
1-Fold	89%	89%	89%	89%
2-Fold	89%	89%	89%	89%
3-Fold	89%	89%	89%	89%
4-Fold	89%	89%	89%	89%
5-Fold	92%	92%	92%	92%

VII. CONCLUSION

This study presents the application of machine learning for the classification of eye diseases, utilizing a dataset comprising retinal images of both normal and diseased eyes. Initially, image processing techniques such as grayscale conversion, HE, and resizing are utilized to enhance image quality and optimize algorithm performance. Then, feature extraction are conducted using HOG, LBP, and ResNet-50 to capture distinctive image characteristics. Finally, KNN, SVM, and XGBoost algorithms are used for disease prediction. The comperative results reveal the superior performance of models trained with image processing and feature extraction. The combination of XGBoost with ResNet-50 achieved the highest performance with 92% accuracy, followed by the combination of SVM with ResNet-50 at 90% accuracy. The study's findings highlight the potential of machine learning in the field of ophthalmology by enabling the development of automated diagnostic tools that can aid in the early detection and treatment of eye diseases, ultimately leading to improved patient outcomes and a reduction in the global burden of visual impairment.

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